

Exploring Race Data from the 2024 Monaco Formula 1 Grand Prix

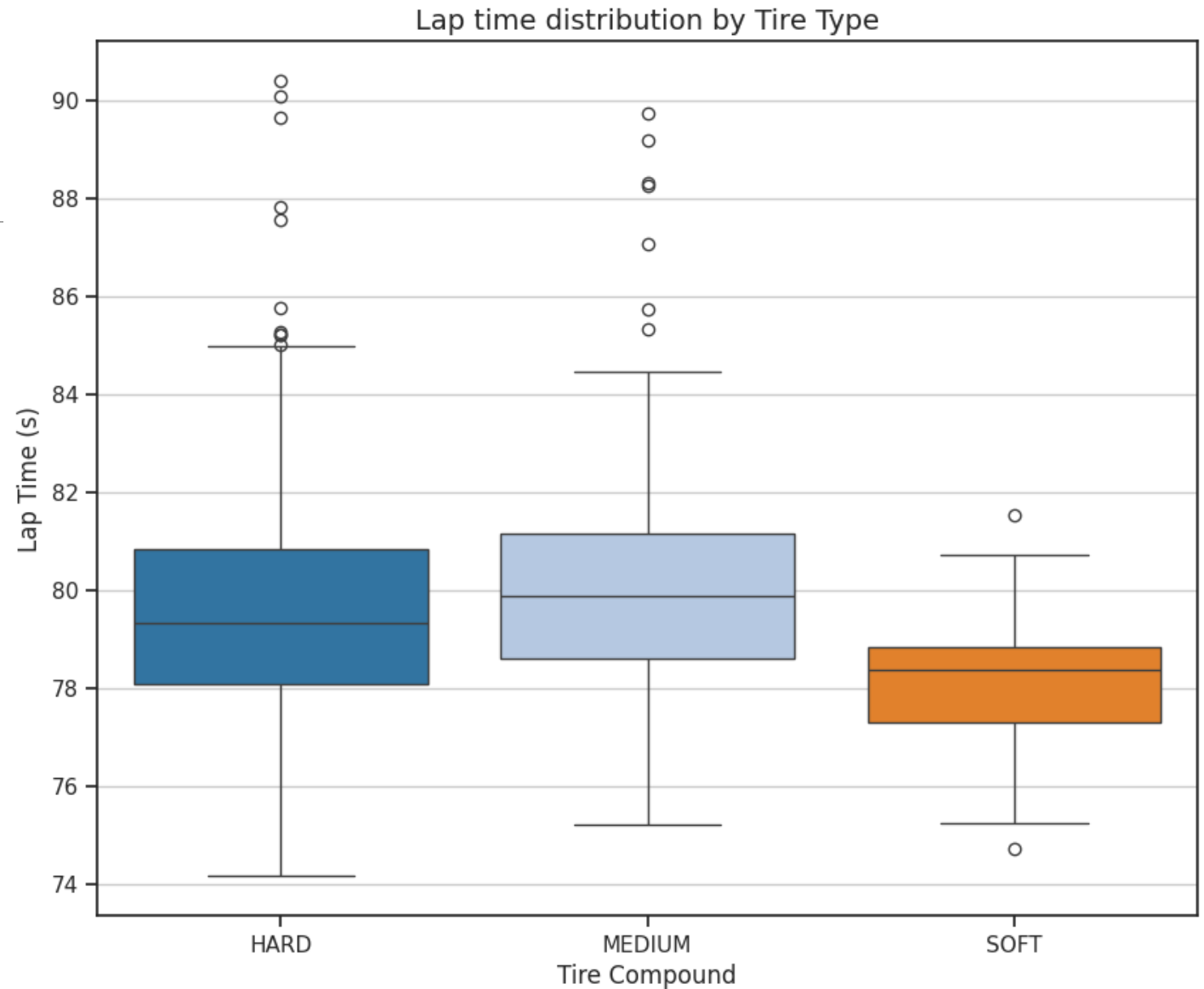
DANIELA MELO, 04/2/2026

Data Overview and Key Variables

Monaco race data extracted from FastF1 python package.

1187 Laps used in analysis

Variable	Description	Type
Driver	Driver's name.	Categorical
Compound	Type of tire used.	Categorical
TyreLife	Age of tire in number of laps.	Numeric
LapNumber	Lap number in Monaco race.	Numeric
lap_time_s	Lap time in seconds.	Numeric



Tire Age and Type As Related to Lap Time

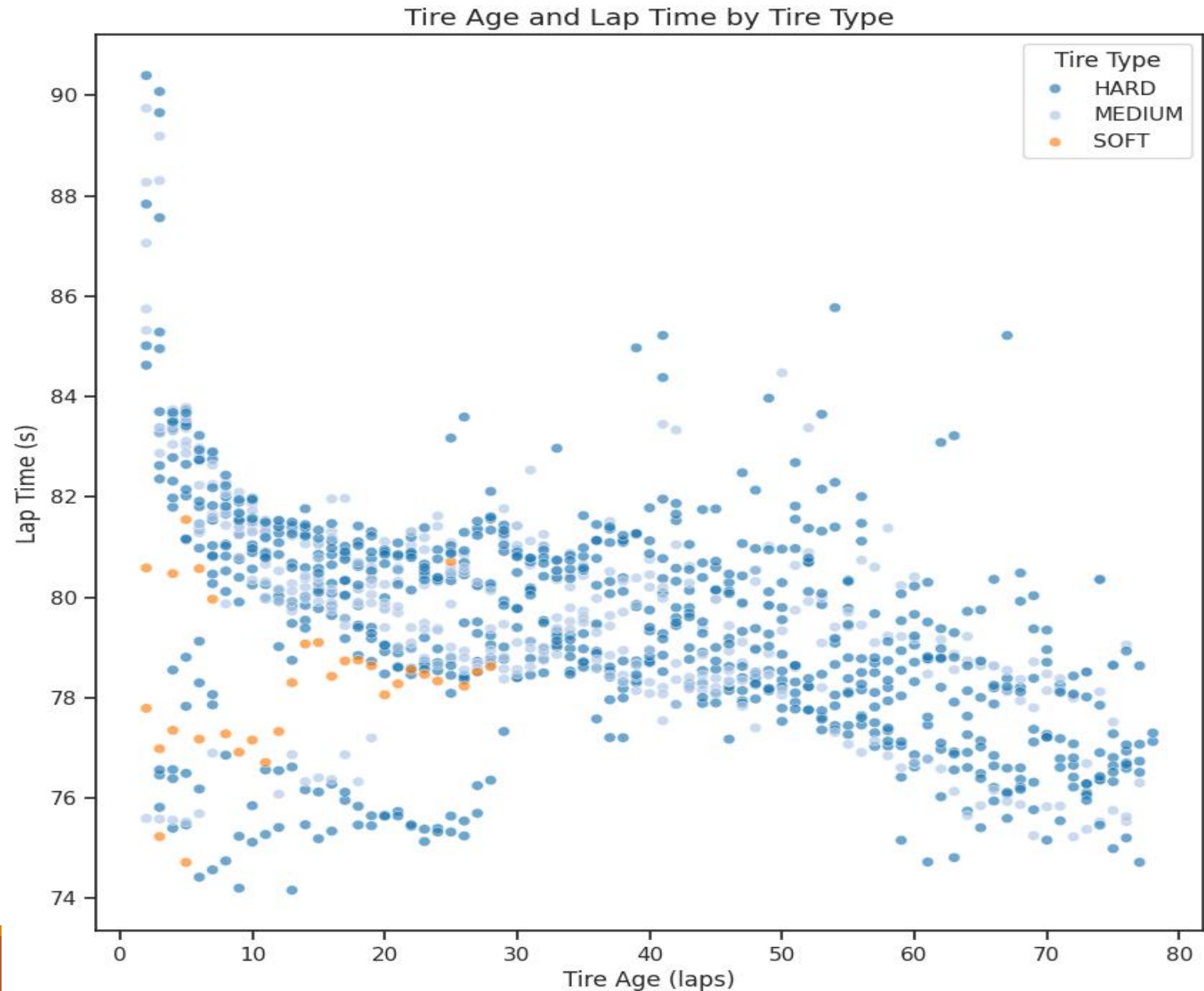
Linear Regression Model:

$\text{lap_time_s} \sim \text{TyreLife} + \text{C(Compound)}$

$R^2 = 0.245$

BUT P-values all below 0.05

	coef	std err	t	P> t
Intercept	81.18	0.119	685.02	0
C(Compound)[MEDIUM]	0.3075	0.119	2.584	0.01
C(Compound)[T.SOFT]	-2.1884	0.349	-6.268	0
TyreLife	-0.0499	0.003	-18.756	0



Bootstrap Estimation of Median Lap Time

Extracted 10% subset of the data.

Subset was resampled 5000 times.

95% CI:

Lower: 79.029

Upper: 80.304

Bootstrap median: 79.74500 s
Full Sample Median: 79.48400 s

